

Report on Free Docker Hosting Platforms for Next.js Applications

This report aims to identify and evaluate Docker hosting platforms that offer genuinely free tiers suitable for deploying Next.js applications. The focus is on platforms that provide a persistent free tier, not just free trials, and that support containerized deployments rather than solely static site hosting. This evaluation considers the necessity of Docker containers for deploying Next.js applications, especially when targeting platforms like AWS, Google Cloud Run, or other cloud providers that may not offer seamless integration like Vercel ([Itsuki, 2023](#)).

Docker and Next.js Deployment

Docker has become increasingly vital for modern application deployment, offering consistency across different environments, from local machines to servers and cloud platforms ([Dhiwise, n.d.](#)). Next.js, a React framework for building server-side rendered and statically generated web applications, greatly benefits from containerization.

There are several reasons for choosing Docker, which are:

- **Consistency:** Packaging Next.js apps and their dependencies into Docker images ensures that the application behaves the same way regardless of the environment ([Dhiwise, n.d.](#)).
- **Efficiency:** Docker containers are lightweight and standalone, containing everything needed to run the software, including code, runtime, libraries, and environment variables ([Dhiwise, n.d.](#)).
- **Scalability:** Docker facilitates scalable and efficient deployment, utilizing the full power of containerization ([Dhiwise, n.d.](#)).

While Next.js can be deployed on platforms like Vercel without Docker, due to Vercel's native support for the framework, Docker becomes essential for deployment on other cloud providers like AWS or Google Cloud Run ([Itsuki, 2023](#)). This is because Docker standardizes the deployment process,

encapsulating the application and its dependencies in a consistent environment.

Evaluation Criteria for Free Docker Hosting Platforms

The primary criterion is the availability of a genuinely free tier that supports Docker container deployment. The evaluation excludes platforms that offer only free trials, focusing instead on services that provide a perpetual free option, albeit potentially with limitations on resources or usage.

Key factors considered:

- **Docker Support:** Must natively support Docker container deployment.
- **Free Tier Availability:** A sustainable free tier, not just a temporary trial.
- **Resource Limits:** Understanding the limitations of the free tier (e.g., CPU, memory, storage, bandwidth).
- **Ease of Use:** How easy it is to deploy and manage containers.
- **Scalability Options:** Even within a free tier, understanding possible upgrade paths.
- **Community and Documentation:** Availability of community support and documentation.

Potentially Free Docker Hosting Platforms (Considerations Needed)

Based on the information provided and general knowledge, determining services offering completely free Docker hosting for Next.js application without the need for credit card information and limited use. However, it's likely that the solutions listed in the previous response are most likely to be best suited for a completely free solution.

Fly.io ([FlyWP, 2025](#))

Fly.io is mentioned as a Docker hosting platform. Its free tier typically provides limited resources (e.g., shared CPUs, limited memory) suitable for small applications or testing.

Koyeb ([Koyeb, n.d.](#))

Koyeb facilitates Docker deployment and offers a serverless platform. The platform could potentially offer a free tier with limitations.

Deployment Strategies and Tools

Beyond evaluating specific platforms, it's crucial to understand the broader ecosystem of tools and strategies for Next.js and Docker deployment.

Dockerfiles for Next.js

Creating an optimized Dockerfile is critical for efficient Next.js deployment. A multi-stage Dockerfile is often recommended to reduce image size and improve build times ([Koyeb, n.d.](#)).

Example of a multi-stage Dockerfile for Next.js:

```
```dockerfile
```

## Install dependencies

```
FROM node:lts as dependencies WORKDIR /my-project COPY package.json
yarn.lock ./ RUN yarn install --frozen-lockfile
```

## Build the Next.js app

```
FROM node:lts as builder WORKDIR /my-project COPY . . COPY --
from=dependencies /my-project/node_modules ./node_modules RUN yarn build
```

## Configure the runtime environment

```
FROM node:lts as runner WORKDIR /my-project ENV NODE_ENV production
COPY --from=builder /my-project/next ./next COPY --from=builder /my-
```

```
project/public ./public COPY package.json . COPY next.config.js . EXPOSE
3000 CMD ["yarn", "start"] ````
```

## Deployment via VPS (Virtual Private Server)

An alternative to fully managed platforms is deploying a Next.js application directly on a VPS. This approach offers more control but requires more technical expertise. The basic steps include ([Penombre, 2025](#)):

1. Installing system dependencies (Node.js, npm).
2. Cloning and setting up the Next.js project.
3. Creating a systemd service for automatic startup.
4. Setting up Nginx as a reverse proxy.
5. Generating an SSL certificate using Certbot.

## Automation with CI/CD

To automate Docker builds and deployments, integrating CI/CD (Continuous Integration/Continuous Deployment) pipelines is crucial ([DevOps.dev, n.d.](#)). Tools like GitHub Actions and GitLab CI can automate the process of building Docker images and deploying them to hosting platforms.

## Conclusion

Selecting a truly free Docker hosting platform for Next.js applications requires careful evaluation of resource limits and service restrictions. While options like Fly.io and Koyeb are often mentioned, their free tiers must be examined individually to ensure suitability for the project's needs.

## References

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